

HAMILTON INTL, ON, Canada

WMO: 712630

Lat: 43.1734N

Lon: 79.9340W

Elev: 780

StdP: 14.29

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.7	(c) 3.3	(d) -11.2	(e) 3.1	(f) -0.9	(g) -5.2	(h) 4.3	(i) 4.3	(j) 30.4	(k) 26.5	(l) 28.5	(m) 26.2	(n) 8.7	(o) 240	(p) 0.590

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	86.9	72.5	84.2	71.1	81.5	69.8	75.1	83.0	73.3	80.9	71.7	78.6	11.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.7	124.6	79.1	70.9	117.2	77.3	69.4	111.1	75.7	39.1	83.3	37.4	81.1	35.9	78.8	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.5	20.7	DB	-8.5	90.9	6.0	3.3	-12.9	93.3	-16.4	95.2	-19.8	97.0	-24.2	99.4
			WB	-9.2	78.2	5.8	2.3	-13.4	79.8	-16.8	81.2	-20.1	82.5	-24.3	84.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	22.7	23.9	32.2	43.8	56.0	65.3	70.2	68.6	62.2	50.1	39.2	28.9	(6)
(7)		DBStd	18.80	11.31	10.22	10.10	8.69	8.31	6.57	5.30	5.12	7.01	8.26	8.45	9.21	(7)
(8)		HDD50	3488	847	732	558	225	32	0	0	0	2	103	334	655	(8)
(9)		HDD65	7024	1311	1152	1018	636	303	74	13	23	135	466	773	1120	(9)
(10)		CDD50	2409	0	0	5	40	217	460	625	576	367	107	11	1	(10)
(11)		CDD65	470	0	0	0	1	23	85	173	134	50	4	0	0	(11)
(12)		CDH74	3555	0	0	2	16	233	695	1341	899	344	25	0	0	(12)
(13)		CDH80	880	0	0	0	0	45	189	372	195	77	2	0	0	(13)
(14)	Wind	WSAvg	9.8	11.5	11.4	11.0	11.5	9.7	8.6	7.6	7.3	8.1	9.6	10.3	11.1	(14)
(15)	Precipitation	PrecAvg	34.5	2.8	2.3	2.5	3.2	3.1	2.8	3.4	2.8	3.1	3.1	2.8	2.6	(15)
(16)		PrecMax	44.5	5.2	4.5	5.5	6.1	6.1	7.6	6.9	6.3	7.6	6.6	5.2	5.9	(16)
(17)		PrecMin	25.2	1.3	1.0	0.9	1.4	1.0	0.8	0.7	1.3	0.8	0.7	0.7	1.1	(17)
(18)		PrecStd	5.0	1.1	0.9	1.3	1.2	1.3	1.6	1.7	1.2	1.4	1.4	1.1	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	54.4	66.8	76.4	84.7	88.5	90.8	89.4	86.8	77.0	63.9	55.1	(19)
(20)			MCWB	52.0	48.5	57.4	62.8	69.9	72.2	74.9	74.5	73.0	67.3	55.8	52.0	(20)
(21)		2%	DB	47.4	46.9	57.6	70.0	80.5	85.3	87.5	84.8	81.9	71.6	59.3	49.5	(21)
(22)			MCWB	45.4	43.8	50.8	57.9	67.2	70.8	73.1	72.4	71.0	63.6	54.4	46.8	(22)
(23)		5%	DB	41.4	41.5	51.8	64.2	75.9	81.7	84.1	81.7	77.9	67.0	56.2	44.7	(23)
(24)			MCWB	39.0	38.4	46.8	54.5	64.3	69.7	70.9	70.7	67.9	60.7	51.4	42.4	(24)
(25)		10%	DB	37.0	36.8	46.6	58.7	71.1	77.7	81.2	79.0	74.2	63.1	52.1	40.7	(25)
(26)			MCWB	35.1	34.3	42.1	50.7	61.1	66.7	69.4	69.2	66.2	57.9	48.3	38.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	50.8	58.6	64.5	72.1	74.7	78.1	77.0	75.3	69.3	58.5	52.2	(27)
(28)			MCD B	54.0	52.2	64.5	74.9	82.2	84.0	86.7	83.3	83.1	75.0	61.1	54.0	(28)
(29)		2%	WB	45.6	44.0	52.6	59.7	68.8	72.5	75.5	74.4	72.5	64.9	55.2	47.4	(29)
(30)			MCD B	46.9	46.8	56.5	67.3	78.1	82.1	83.3	81.8	79.0	69.2	58.8	48.9	(30)
(31)		5%	WB	39.3	38.5	47.1	56.0	66.0	70.7	73.5	72.6	70.1	61.6	51.9	42.7	(31)
(32)			MCD B	40.9	40.9	51.3	62.6	73.6	79.5	81.2	79.0	75.4	66.0	55.6	44.5	(32)
(33)		10%	WB	35.3	34.3	42.0	51.0	62.8	68.6	71.7	70.9	67.8	58.7	48.7	38.7	(33)
(34)			MCD B	36.8	36.4	46.0	57.6	69.4	75.6	78.1	76.6	72.7	62.1	52.0	40.4	(34)
(35)	Mean Daily Temperature Range	MDBR		13.0	13.7	14.8	17.2	19.1	18.2	18.6	17.9	17.9	15.4	13.1	11.1	(35)
(36)		5% DB	MCD BR	15.7	18.3	22.7	27.1	25.0	22.5	21.6	20.7	22.0	20.5	18.8	15.0	(36)
(37)			MCWBR	14.8	16.2	17.0	18.0	14.8	12.0	10.8	10.8	12.0	13.2	14.7	13.6	(37)
(38)		5% WB	MCD BR	14.8	17.6	22.0	24.5	22.8	20.5	19.3	18.0	19.5	18.5	17.3	14.9	(38)
(39)			MCWBR	14.5	16.2	17.6	18.3	14.6	12.0	10.6	10.4	12.1	13.1	14.5	14.0	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.289	0.332	0.330	0.374	0.396	0.432	0.425	0.410	0.377	0.357	0.307	0.291	(40)
(41)		taud		2.401	2.250	2.401	2.314	2.302	2.224	2.266	2.294	2.370	2.413	2.520	2.461	(41)
(42)		Ebn at Noon		269	271	289	283	279	267	268	268	269	260	262	259	(42)
(43)		Edh at Noon		24	34	33	39	41	44	42	39	34	29	22	21	(43)
(44)	All-Sky Solar Radiation	RadAvg		446	685	1075	1410	1721	1889	1886	1658	1321	799	497	334	(44)
(45)		RadStd		44	73	119	153	174	132	135	86	123	75	68	39	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	+1.01	N/A	N/A	N/A	N/A
(47) Regional (54 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page